

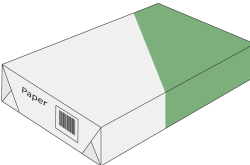
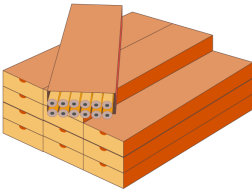
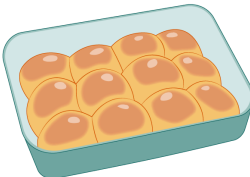
The mole and Avogadro constant

Learning outcomes

By the end of this section you should be able to:

- Explain the use of the term mole to represent the amount of elementary entities, which includes atoms, molecules and formula units.
- Calculate amounts of substance n (given in moles) to the total number of specified elementary entities (atoms, molecules, ions and electrons).

Different units are used to count groups of items: a pair means two items, a dozen contains 12 items, a gross contains 144 items and a ream contains 500 items (**Interactive 1**). A pair will always represent two items, whether it represents the number of eggs, test tubes or people. Counting a large quantity of items arranged into groups allows for the process to be manageable, which is more practical for counting such large quantities.



ream

pair

dozen

gross

☒ Check

Interactive 1. Match the Unit of Measurement for Each of These Objects.

You may have noticed above that the smaller the object, the greater the number of items that are grouped together. For instance, a pair of gloves or shoes containing two larger items compared to a ream of paper containing five hundred thin sheets of paper. Can you imagine the number of tiny atoms that might be grouped together?

The mole (mol) is the unit used to measure the amount of substance (n). One mole contains exactly the number of elementary entities given by the Avogadro constant (or N_A) (**Figure 1**). It was named after the Italian scientist Amedeo Avogadro, and its value is given in [section 2](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/physical-constants-id-45104/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/physical-constants-id-45104/>) of the DP chemistry data booklet as $6.02 \times 10^{23} \text{ mol}^{-1}$. One mole is used to represent the number of elementary entities, but it could also be used to represent 6.02×10^{23} of anything.

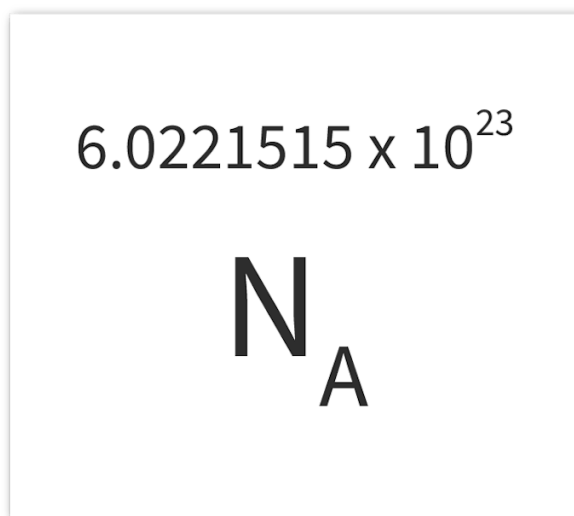

$$6.0221515 \times 10^{23}$$
$$N_A$$

Figure 1. The Avogadro constant. For calculations, this number is rounded to 6.02×10^{23} .

 More information for figure 1

The image displays the notation for Avogadro's constant. At the top, there is a number written in scientific notation as " 6.0221515×10^{23} ". Below this, centered in the image, is the letter "N" with a subscript "A" (N_A), which denotes Avogadro's number. The image is enclosed within a rectangular border, providing a neat presentation of the information.

[Generated by AI]

Do you think you now understand what a mole represents? Can you create an image in your mind of how big this number is? Watch **Video 1** to dig deeper into the concept of the mole and to put the magnitude of the Avogadro constant into perspective.

Introduction to Moles



Video 1. An introduction to moles.

Recall from section S1.1.1a (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/elements-compounds-and-mixtures-id-43065/>) that pure substances are classified as elements and compounds. In this section, we are going to introduce a new unit, named the mole, that represents an amount of chemical particles, known as elementary entities (see **Table 1**). Elementary entities can be atoms, molecules, ions, electrons or any other chemical particle, such as a formula unit. A formula unit represents the simplest ratio of ions in an ionic compound, for example sodium and chloride ions bond in a 1:1 ratio, making the formula NaCl. You will learn more about formula units in section S2.1.2 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ionic-bonding-id-43557/>).

Table 1. The number of particles in one mole of some elementary entities.

Elementary entity	Type of matter	Example	Number of particles
1 mol of atoms	element	Fe	6.02×10^{23} atoms
1 mole of molecules	molecular compound	H ₂ O	6.02×10^{23} molecules
1 mole of ions	ion	Cl [−]	6.02×10^{23} ions
1 mole of electrons	subatomic particle	e [−]	6.02×10^{23} electrons

Nature of Science

Aspect: Measurement

The Avogadro constant is a very large number. But it is a good example of a great tool that science has developed to help us measure matter at the atomic scale.

How big is a mole? (Not the animal, the other one.) - Da...



Video 2. How big is a mole?

Study skills

One mole of any substance contains 6.02×10^{23} elementary entities of that substance. This number is given in the DP chemistry data booklet so you don't need to memorise it, but you will have to be confident in using it in your calculations and conversions. Be sure that you are correctly using the exponent functions on your calculator for these calculations.

We use the Avogadro constant to calculate the number of atoms, molecules, ions or electrons. Since the unit for the Avogadro constant is mol^{-1} , this quantity allows for the units to vary, depending on the type of elementary entity being calculated (**Table 2**).

Table 2. Units of different elementary entities.

Elementary entity	Avogadro constant
atoms	$6.02 \times 10^{23} \text{ atoms mol}^{-1}$
molecules	$6.02 \times 10^{23} \text{ molecules mol}^{-1}$
ions	$6.02 \times 10^{23} \text{ ions mol}^{-1}$
electrons	$6.02 \times 10^{23} \text{ electrons mol}^{-1}$

Theory of Knowledge

The magnitude of the Avogadro constant of 6.02×10^{23} is well beyond the scale of quantities of objects in our daily lives. Even the number of stars in the Milky Way galaxy is estimated to be only 100–400 billion stars. To what extent does our ability to comprehend this magnitude limit our understanding of the quantity of elementary entities in 1 mole?

Let's do a few examples using the Avogadro constant.

Worked example 1

Calculate the number of atoms in 1.00 mole of carbon.

Solution steps	Calculations
Step 1: recognise that the Avogadro constant will be required.	The question involves finding a given number of moles so the Avogadro constant will be required.
Step 2: write out the values given in the question. The value for the Avogadro constant is in the DP chemistry data booklet.	Amount of carbon = 1.00 mol Avogadro constant = 6.02×10^{23} mol ⁻¹
Step 3: Identify the equation used to find the number of atoms.	Number of atoms = number of moles $\times$ Avogadro constant
Step 4: substitute the values into the formula.	Number of atoms = 1.00 mol $\times$ 6.02×10^{23} mol ⁻¹
Step 5: state the answer with the appropriate number of significant figures used in rounding.	= 6.02×10^{23} atoms of carbon

Worked example 2

Calculate the number of moles in 1.20×10^{24} atoms of iron.

Solution steps	Calculations
Step 1: recognise that Avogadro's constant will be required.	The question involves finding a given number of carbon constant will be required.
Step 2: write out the values given in the question. The value for the Avogadro constant is in the DP chemistry data booklet.	Amount of iron = 1.20×10^2 Avogadro constant = 6.02×10^{23}
Step 3: Identify the equation used to find the number of atoms.	Number of atoms = $\frac{\text{number of moles} \times \text{Avogadro constant}}{\text{atoms per mole}}$
Step 4: make the number of moles the subject of the formula.	Number of moles = $\frac{\text{number of atoms}}{\text{Avogadro constant}}$
Step 5: substitute the values into the formula.	Moles of Fe = $\frac{1.20 \times 10^2}{6.02 \times 10^{23}}$ $= 1.9934 \times 10^{-22}$
Step 6: state the answer with the appropriate number of significant figures used in rounding.	$= 1.99$ (to 3 s.f.)

Worked example 3

Determine the number of water molecules in 2.50 moles of water.

Solution steps	Calculations
Step 1: recognise that the Avogadro constant will be required.	The question involves finding a given number of moles so the Avogadro constant will be required.
Step 2: write out the values given in the question.	Amount of water = 2.50 moles Avogadro constant = 6.02×10^{23}

Solution steps	Calculations
The value for the Avogadro constant is in the DP chemistry data booklet.	
Step 3: Identify the equation used to find the number of molecules.	Number of molecules = $n \times N_A$
Step 4: substitute the values into the formula.	Number of molecules = 2.51×10^{24}
Step 5: state the answer with the appropriate number of significant figures used in rounding.	$= 1.51 \times 10^{24}$ molecules of CH_4

Worked example 4

Calculate the number of electrons in 3.00 moles of carbon, C.

Solution steps	Calculations
Step 1: recognise that the Avogadro constant will be required.	The question involves finding the given number of moles so the Avogadro constant will be required.
Step 2: write out the values given in the question. The value for the Avogadro constant is in the DP chemistry data booklet.	Amount of carbon = 3.00 mol Avogadro constant = 6.02 × 10 ²³ mol ⁻¹ Number of electrons in 1 mol of carbon = 6 × 10 ²³
Step 3: Identify the equation used to find the number of electrons.	Number of electrons = 6 × 10 ²³ × 3.00 mol = 6 × 10 ²³ × 3.00 = 1.8 × 10 ²⁴
Step 4: substitute the values into the formula	Number of atoms = 6 × 10 ²³ × 3.00
Step 5: state the answer with the appropriate number of significant figures used in rounding.	= 1.08 × 10 ²⁵ electrons

Activity

- **IB learner profile attribute:** Inquirers
- **Approaches to learning:** Thinking skills — Combining different ideas in order to create new understandings
- **Time required to complete activity:** 20 minutes
- **Activity type:** Pair activity

Materials required: 10 coins, blocks or other small items that can be stacked on top of one another, a ruler and pencil and paper.

1. With a partner, make a tower of the ten items and precisely measure the height with a ruler to the nearest millimetre. Calculate how high your tower would be, in kilometres, if you used 1 mole of those items. Express your answer with the correct number of significant digits. (Optional: measure the uncertainty of the ruler and propagate the uncertainty with your calculations.)
2. Using a stopwatch, measure the length of time required to do 20 jumping jacks in seconds. Calculate how many years it would take to do 1 mole of jumping jacks. Express your answer to the correct number of significant figures.

digits. (Optional: measure the uncertainty of the stopwatch and propagate the uncertainty with your calculations)

The number of stars in the Milky Way galaxy is estimated to have at least 100 billion stars. Calculate the amount, in mol, of stars in the Milky Way.